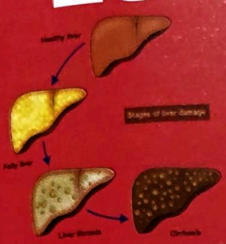
CHAPTER

28



Hepatobiliary Diseases

Learning Objectives

- To have a brief introduction of hepato biliary diseases
- To know the clinical features of liver cirrhosis, alcoholic liver disease (ALD), and non alcoholic liver disease (NALD), hepatitis, jaundice, ascites
- To know about the necessary physical examinations to be performed in above said diseases
- To know about the investigations to be ordered in the confirmation of above said diseases
- To know about the differential diagnosis of above said diseases

Introduction

Hepatobiliary system comprises liver and gall bladder and conducting ducts for the transport of bile from the liver to duodenum. Liver is one of the major organ which maintains most of the activities of human body in terms of metabolism, storage, formation of plasma proteins, excretion of metabolites etc. Depending upon the nature of aetiological exposure, the hepatobiliary system may show abnormal functions which can be learnt by observing specific clinical features which are pertaining to the hepatobiliary system such as icterus (yellowish discolouration of skin, sclera, tongue and mucous membranes) or pain in right upper quadrant or haematemesis or malena or ascites or certain signs such as spider naevus or palmar erythema or marks of itching on the skin with yellowish discolouration. As hepatobiliary system will be functioning coherently with other systemic bodily functions, this system will also be afflicted with pathological events like other bodily systems which show symptoms such as fever, weight loss etc. Diagnosis of these dysfunctions can be done by CBC, liver function tests (LFT), imaging studies. The following chapter will be discussing some of the important pathological processes happening in and around hepatobiliary system.

Cirrhosis of Liver

Before discussing the cirrhosis **(see Figure 84)** of liver, we have to understand the term fibrosis of liver. Hepatic fibrosis is overly exuberant wound healing in which excessive connective tissue builds up in the liver. The extracellular matrix is overproduced, degraded deficiently or both. The trigger is chronic injury to the hepatic tissue. Fibrosis itself causes no symptoms but can lead to portal hypertension or cirrhosis results in disruption of normal hepatic architecture and liver dysfunction. Diagnosis is done by liver biopsy.

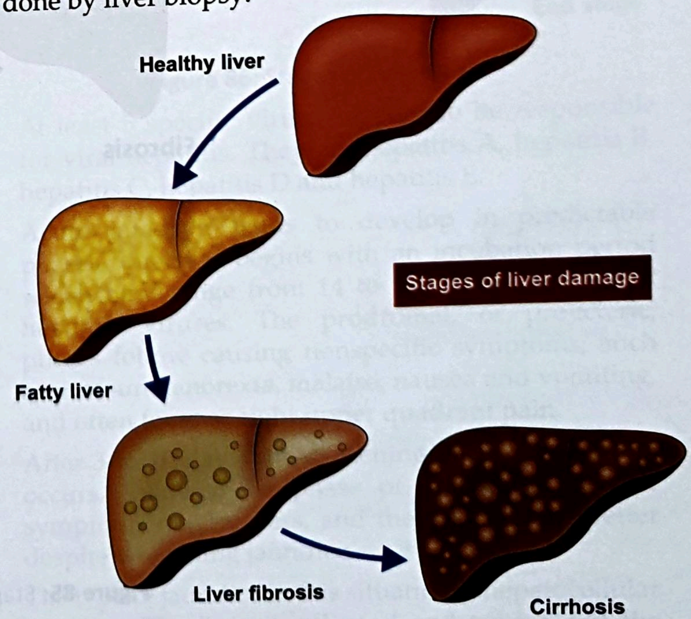


Figure 84: Stages of liver Damage

Cirrhosis is a late stage of fibrosis of the liver which resulted in widespread distortion of normal hepatic structural design. Cirrhosis is characterized by regenerative nodules surrounded by dense fibrotic tissue. Symptoms may not develop for years and are often nonspecific (e.g. anorexia, weight loss, fatigue). Late manifestations include portal hypertension, ascites and liver failure. Diagnosis often requires liver biopsy, while prediction of diagnosis can be inferred by ultrasound abdomen.

- Cirrhosis may be asymptomatic for years. 1/3rd of the patients may show symptoms. Often, first symptoms are non specific, they include general fatigue, anorexia, malaise, weight loss.
- Liver is palpable and firm with blunt edge, nodules are not usually palpable.
- Clinical signs may suggest a chronic liver disorder or chronic alcohol use, but are not specific for cirrhosis include muscle wasting, palmar erythema, parotid gland enlargement, clubbing, spider naevus.
- Diagnosis[1] is done by liver function tests, coagulation tests, CBC, biopsy of the liver
- Prognosis is often unpredictable. It depends on factors such as aetiology, severity, presence of complications, comorbid conditions and effectiveness of therapy.

ALD and NALD

Alcoholic Liver Disease

Alcoholic Liver Disease (ALD) (see Figure 85) is a type of hepatic dysfunction which occurs due to the excess consumption of ethyl alcohol by an individual. Disorders that occur in alcohol abusers, often in sequence include.

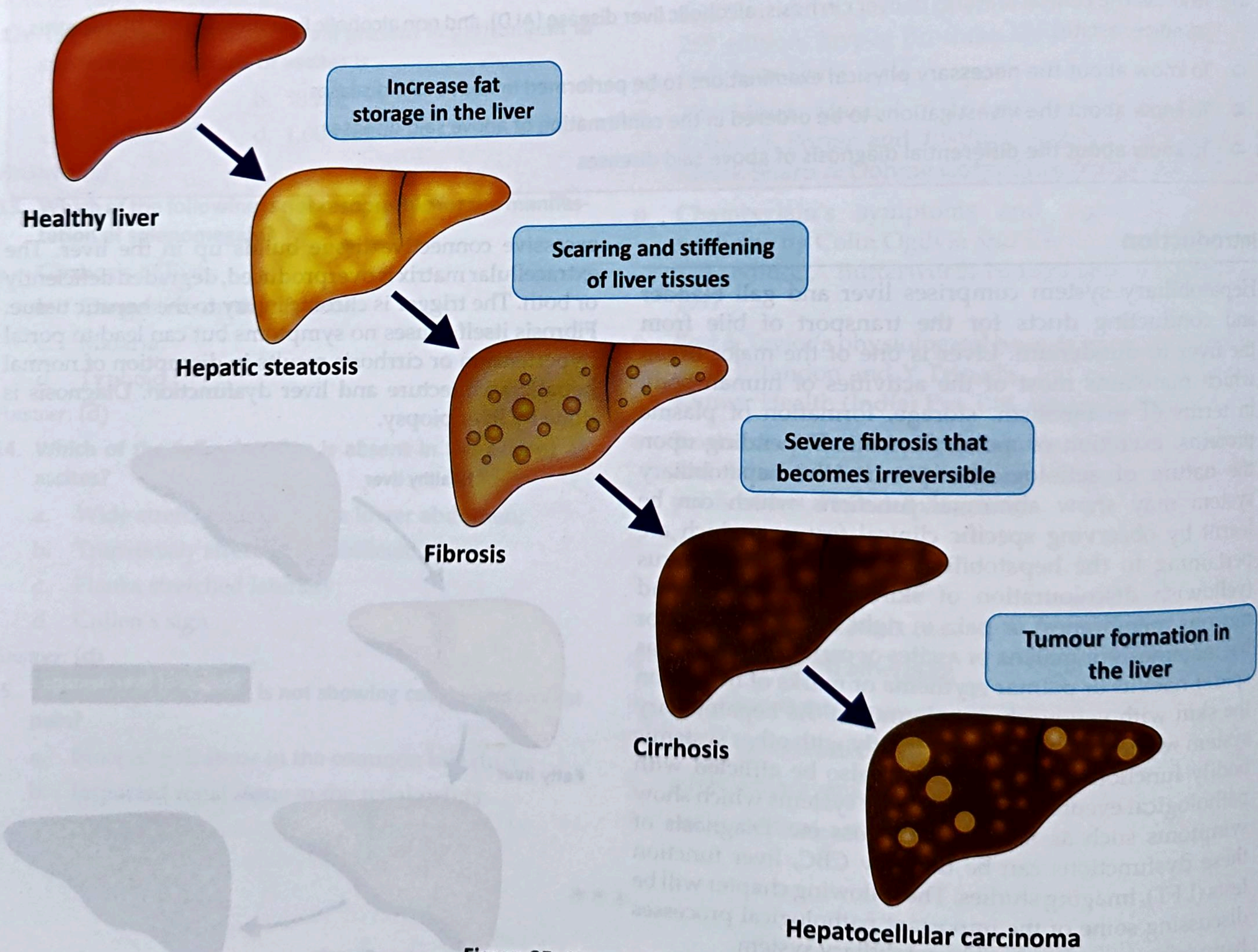


Figure 85: Stages of liver damage

1 SGOT and SGPT levels may moderately increased; elevated alkaline phosphatase level may imply biliary obstruction; Bilirubin is usually normal, but increases when cirrhosis becomes progressive; decreased serum albumin and prolonged PT time may help in the diagnosis of impaired hepatic synthesis. Anemia may be multifactorial either due to GI bleeding or due to deficient vitamin B12 or folic acid or due to hemolysis or due to hypersplenism

- Fatty liver (in > 90%)
- Alcoholic hepatitis (in 10 to 35%)
- Cirrhosis (in 10 to 20%)

Non alcoholic liver disease

Non Alcoholic Liver Disease (NALD) is the term used for a range of conditions caused by a build-up of fat in the liver. This is usually seen in people who are overweight or obese. Early stage NALD does not cause any harm usually, but it can lead to serious liver damage, including cirrhosis, if it gets worse.

- Fatty liver, alcoholic hepatitis and cirrhosis of liver may overlap in the individuals. For better understanding, this is dealt separately in brief.
  - **Fatty liver (steatosis):** is the initial and most common consequence of excessive alcohol consumption. Fatty liver is potentially reversible. Macrovascular fat accumulates as large droplets of triglycerides and displaces the hepatocyte nucleus, most markedly in perivenular hepatocytes. This leads to enlargement of liver.
  - **Alcoholic hepatitis:** This is the combination of fatty liver and hepatic inflammation and necrosis of liver. This may vary depending upon the severity. Damaged hepatocytes become swollen. Severely damaged hepatocytes get necrosed leading to the narrowing of sinusoids and terminal hepatic venules.
  - **Cirrhosis:** There will extensive fibrosis that disturbs the hepatic structure. The amount of fat varies, alcoholic hepatitis may coexist. The hepatic regeneration produces relatively smaller nodules leading the shrinking of liver.
  - **Fatty liver is usually asymptomatic,** enlargement observed in this situation is non tender. In case of alcoholic hepatitis, the person may have fatigue, fever, jaundice, right upper quadrant pain, tender hepatomegaly. In cirrhosis, the person may have portal hypertension where in there may be the possibility of GI bleeding (haemetemesis or malena), ascites, portal systemic encephalopathy.
- In case of NALD, the above features may be observed with the negative history of ethyl alcohol intake.
- Diagnosis is done by analysing the history of alcohol intake, liver function tests and CBC.
- Prognosis is determined by the degree of hepatic fibrosis and inflammation. Fatty liver and alcoholic hepatitis without fibrosis are reversible, if alcohol is avoided. With abstinence from alcohol, fatty liver is completely reversible within 6 weeks. Fibrosis and cirrhosis phases of liver disease are irreversible.

Hepatitis

Hepatitis **(see Figure 86)** is the inflammation of liver characterized by diffuse or patchy necrosis. Major causes are septic hepatic viruses, alcohol and drugs. In this section, most commonly occurring hepatitis due to hepatitis viruses is being discussed. Acute viral hepatitis is diffuse inflammation of liver caused by specific hepatotropic viruses. This is characterized by anorexia, nausea and often fever or right upper quadrant pain. Jaundice often develops, as other symptoms begin to resolve. Most cases resolve spontaneously but some progress to chronic hepatitis. Occasionally, acute viral hepatitis progresses to acute hepatic failure (fulminant hepatitis). Diagnosis is done by liver function tests and serologic tests to identify viruses.

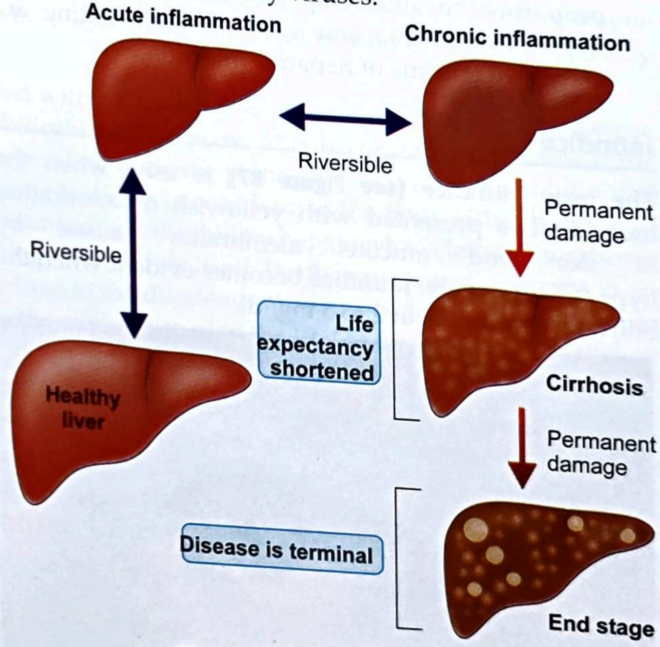


Figure 86: Stages of hepatitis

- At least 5 specific viruses appear to be responsible for viral hepatitis. They are hepatitis A, hepatitis B, hepatitis C, hepatitis D and hepatitis E.
- Acute infection tends to develop in predictable phases. Infection begins with an incubation period which may range from 14 to 180 days in different hepatitis viruses. The prodromal, or pre-icteric, phases follow causing nonspecific symptoms, such as profound anorexia, malaise, nausea and vomiting, and often fever or right upper quadrant pain.
- After 3 to 10 days, the darkening of colour of urine occurs, followed by phase of jaundice. Systemic symptoms may regress, and the patient feels better despite worsening jaundice.
- The type of jaundice in this situation is hepatocellular jaundice. The liver is enlarged and tender, but the edge of the liver remains soft and smooth. Mild splenomegaly occurs in 20% of the patients.

- Jaundice usually peaks within 2 to 3 weeks and then fades during a 2 to week to get recovered. Acute viral hepatitis usually resolved spontaneously 4 to 8 hours after symptom onset.
- As far as physical examination of the patient is concerned, the person will have yellowish discolouration of the ventral surface of tongue, sclera, skin and oral mucosa. On inspection, the person may have tenderness of right upper quadrant; signs of itching with marks of itching due to the deposition of bile pigment, signs of ascites in case of chronic hepatitis are seen.
- Diagnosis is done by liver function test (AST and ALT elevation out of proportion out of proportion to alkaline phosphatase); usually with hyperbilirubinemia and viral serologic testing are usually diagnostic of hepatitis infection.

Jaundice

The term jaundice **(see Figure 87)** is used when the individual is presented with yellowish discolouration of skin and mucous membranes caused by hyperbilirubinemia. Jaundice becomes evident when the bilirubin level is about 2 to 3 mg/ dl.

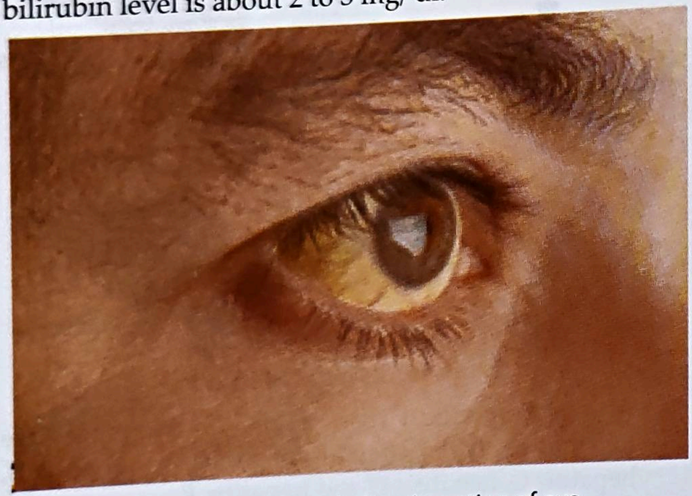


Figure 87: Yellowish discolouration of eye

- Most bilirubin is produced when Hb is broken down into unconjugated bilirubin. Unconjugated bilirubin binds to albumin in the blood for transport to the liver, where it is taken up by hepatocytes and conjugated with glucuronic acid to make it conjugated bilirubin which is water soluble.
- Conjugated bilirubin is excreted in the bile to the duodenum. In the intestine, bacteria metabolise bilirubin to form urobilinogen.
- Some urobilinogen is eliminated in the faecal matter and some other amount of urobilinogen is getting reabsorbed in the ileum and enters back into the liver. The process is termed as enterohepatic circulation.
- **Hyperbilirubinemia may involve**
  - Increased production of bilirubin
  - Decreased hepatic uptake
  - Decreased conjugation
  - Dysfunctional hepatocytes
  - Decreased expulsion of bilirubin from the liver to the gall bladder
  - Decreased expulsion of bilirubin from the gall bladder to the duodenum

Major causes of jaundice

- Prehepatic causes (causes which are responsible for elevation of unconjugated bilirubin): haemolysis due to malaria or congenital spherocytosis or congenital elliptocytosis or sickle cell disease etc
- Hepatic causes such as viral hepatitis; alcoholic liver disease; hepatoma etc
- Post hepatic causes such as cholelithiasis; common bile duct obstruction; carcinoma of head of pancreas; stricture of common bile duct etc
- History of present illness should include onset and duration of jaundice.
- Hyperbilirubinemia can lead to darkening of urine before jaundice is visible.
- Therefore the onset of dark urine indicates onset of hyperbilirubinemia, more accurately than the onset of jaundice. Fever, malaise, myalgia, darkening of urine and faecal matter may appear in the body.
- Pruritus, skin itching marks and steatorrhea (passage of fat in the feacal matter) and right upper quadrant pain.
- Vital signs are review for assessment of fever, systemic toxicity (hypotension and tachycardia) has to be analysed.
- Head and neck examination includes assessment of eyes, tongue and oral mucosa.
- Abdomen is examined for the assessment of liver, ascites.

Diagnostic approach of jaundice

- Acute jaundice in young and healthy suggests acute viral hepatitis
- Acute jaundice appears after acute drug or exposure to toxic substances
- A long history of heavy alcohol use suggests alcoholic liver disease
- A personal or family history of recurrent, mild jaundice without findings of hepatobiliary obstruction, may imply hereditary disorder
- Gradual onset with pruritus, weight loss and clay coloured stools imply intrahepatic or extrahepatic obstruction
- Painless jaundice in elderly patients with weight loss and a mass in abdominal region may imply malignancy of head of pancreas.
- Diagnosis of the disease: liver function tests; imaging technques

Key summary of the chapter

- Hepatobiliary system comprises of liver, gall bladder and vessels which help to move bile into duodenum and blood from portal vein to inferior vena cava to hepatic vein. Liver is one of the major organ which performs more than 70 different activities of the human body related to metabolism, storage, excretion etc. These different functions will get disturbed depending upon the nature of exposure of the aetiological factor, physical strength of the human being.
- Cirrhosis of the liver is a late stage of fibrosis of the liver which resulted in widespread distortion of hepatic structural design. Fibrosis is wound healing occurring in the liver where excessive connective tissue builds up in the liver.
- ALD is the hepatic dysfunction which occurs due to the excess consumption of alcohol. NALD is the hepatic dysfunction occurs in the similar fashion as in ALD, but with negative history of alcohol intake. The pathological events occur in this disease process are fatty liver, alcoholic hepatitis and cirrhosis of liver.
- Hepatitis is the inflammation of liver characterized by diffuse or patchy necrosis. Major causes of this disease are septic hepatic viruses, alcohol and drugs. 5 different species of hepatitis virus are responsible for the manifestation of viral hepatitis. These species are hepatitis A, B, C, D and E. Fever, yellowish discolouration of skin, sclera, tongue and mucous membrane and pain in the right upper quadrant are some of the major clinical features observing during the progression of the disease.
- Jaundice is the term used when the individual is presented with yellowish discolouration of skin, sclera, tongue and mucous membranes due to excess concentration of bilirubin in the body. This is clinically detectable if the serum bilirubin concentration raises above 2 – 3 mg/ dl. Depending upon the pathogenesis, jaundice is subdivided into prehepatic, hepatic and posthepatic jaundice. Prehepatic jaundice is occurring in the body either because of excess haemolysis or because of the deficient uptake of unconjugated bilirubin by hepatocytes. Hepatic jaundice is presented in an individual when he is having main pathology in the liver itself. Posthepatic jaundice occurs in an individual when the course of movement of bile from the liver to the duodenum is disturbed. Each of these types of jaundice will show typical clinical features which helps the clinician to plan the line of treatment in an efficient manner.
- Ascites is the term used when there is excess accumulation of fluid in the peritoneal cavity happens. This occurs mainly because of portal hypertension. The main clinical presentation is excess abdominal girth which makes the person to wear the clothes with difficulty. When the person is having ascites because of hepatic dysfunction, this will be associated with yellowish discolouration of the skin, sclera and mucous membranes and when it occurs because of cardiac dysfunction, this may be associated with the features of cardiac failure.

Self-Assessment: Questions for Reviews

Short essay questions

- Explain alcoholic liver disease
- Describe jaundice
- Describe ascites
- Describe viral hepatitis
- Explain cirrhosis of liver

MCQs

1. **Which of the following is an example of prehepatic jaundice?**
   a. Viral hepatitis;
   b. Malaria;
   c. Carcinoma of head of pancreas;
   d. Stricture of common bile duct

   Answer: (b)

2. **Which of the following is not an example of prehepatic jaundice?**
   a. Sickle cell disease;
   b. Malaria;
   c. Congenital spherocytosis;
   d. Cirrhosis of liver

   Answer: (d)

3. **Which of the following sign is indicative of the diagnosis of ascites?**
   a. Puddle sign;
   b. Murphy's sign;
   c. Cullen's sign;
   d. Trendelenberg sign

   Answer: (a)

4. **Smiling umbilicus is observed in which of the following disease?**
   a. Large ovarian cyst;
   b. Huge ascites;
   c. Acute appendicitis;
   d. Umbilical hernia

   Answer: (b)

5. Viral hepatitis is an example of ....type of jaundice.

a. Prehepatic;

b. Hepatic;

c. Posthepatic;

d. All of the above

Answer: (b)

6. Borborygmi is associated with which of the following condition?

a. Ascites;

b. Large intestinal obstruction;

c. Viral hepatitis;

d. Pregnancy

Answer: (b)

7. Borborygmi is ....type of physical examination.

a. Inspection; b. Palpation;

c. Percussion; d. Auscultation

Answer: (d)

8. The presenting complaint, malena (dark tarry stools) is indicative of site of the lesion in

a. Stomach;

b. Small intestine;

c. Caecum;

d. Large intestine

Answer: (a)

9. Unconjugated bilirubin combines with ....to become conjugated bilirubin.

a. Glucuronic acid;

b. Hydrochloric acid;

c. Carbonic acid;

d. Lactic acid

Answer: (a)

10. In which part of the body, yellowish discolouration is seen in the patient of jaundice?

a. Eye; b. Tongue;

c. Skin; d. Nail

Answer: (b)

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Chapter resources

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